Gauge supply, global propagation, collapse, and singular realization

24 September 2026. Current research entry after compact realization. The four proof notes below address different restrictions of that result. They are not four independent claims of world-first discovery, nor a partition of all Yau26 inputs. Their common question is which information lets an approximate Einstein metric lead to a conclusion about its original smooth manifold.

1. Gauge input need not live on the manifold carrying the output

The finite-cover theorem allows actual monopole inputs on a finite cover $N \to M$, with no bound on its degree d. Under bounded base diameter, a per-sheet projection floor $p/d \geqslant A_0$, and bilateral saturation, it produces an exact metric on the **base** with universal cover $\mathbb{H}^2 \times \mathbb{H}^2$. For a nonorientable base, one actual input on an oriented finite cover suffices. At the Miyaoka–Yau numerical boundary, one actual input with $p \geqslant d(2\chi + 3\tau)(M)$ gives a locally complex-hyperbolic metric directly from small pointwise Ricci defect.

The mechanism is to descend the scalar distance of the Weyl operator from its equality orbit, not the auxiliary form or Chern class. Degree cancels in the normalized estimates. Compactness and smooth-type preservation are then proved downstairs. There is no assertion that an arbitrary Einstein metric found upstairs is invariant and descends.

A concrete diagonal quotient of two even-genus hyperbolic surfaces has $\chi + \tau \equiv 2 \pmod 4$ and hence no almost-complex structure in either orientation, but has the required cover inputs and an exact product quotient metric. This is a classical equality example, not a new manifold. It demonstrates why allowing cover supply genuinely enlarges the input description, and why the conclusion is Riemannian rather than global KE.

2. Large volume no longer forces a small total residual

The preceding no-diameter repair theorem required small unnormalized L^2 Ricci residual. The uniformly local repair theorem replaces that condition by an explicit propagation estimate. If the full gauged Einstein operator has **positive quadratic-form coercivity** λ, and the actual manifold satisfies

$$\operatorname{Vol} B_R(x) \leqslant C_G(1+R)^d e^{HR}, \qquad H^2 < 8\lambda,$$

then an exponentially weighted energy identity gives

$$\|L_g^{-1} f\|_{C^{2,\alpha}_{\mathrm{uloc}}} \leqslant C \|f\|_{C^{0,\alpha}_{\mathrm{uloc}}}.$$

With the stated local regularity hypotheses, pointwise small Ricci defect therefore gives an exact negative Einstein metric, without diameter, volume or global-residual smallness. The proof retains every tensor mode.

Fixed cyclic covers of a closed hyperbolic manifold have uniform linear ball growth and Koiso coercivity. Independent small perturbations in each cell can have total residual growing like $t\sqrt{m}$, yet satisfy the uniform repair criterion. These manifolds already admit Einstein metrics; the increment is a uniform repair neighbourhood for distributed errors. The bounded-chart corollary states the analytic argument in every dimension $n \geqslant 3$, including the cyclic-cover examples and their dimension-dependent coercivity constants. The four-dimensional no-bubble theorem is one supplier of its local charts, not part of the weighted inverse identity itself.

3. Genuine collapse now has an actual-form connection

The collapsed bulk theorem uses maximal local nilpotency rank in the Euclidean-model regime of Naber–Zhang. For bilateral gauge deficit $K = 2r + q_+ + q_- \to 0$ and pointwise Ricci defect tending to zero, local universal-cover balls approach $\mathbb{H}^2 \times \mathbb{H}^2$ outside an exceptional set with

$$\mathrm{Vol}(\mathscr{E})/V \leqslant CK^{1/4}.$$

There is no diameter, base-ball volume, homotopy or total-volume bound. The essential transfer is an estimate for **local averages** whose constant is independent of covering multiplicity. It is false for unnormalized integrals, as an explicit pinching-surface product shows.

This changes the local input regime to include genuine collapse. It does not give a global repair theorem: maximal rank is not automatic, the small exceptional set can carry topology, and transition maps and low modes are not controlled by its volume alone.

4. With nonzero second homotopy, allow the singular limit

The orbifold and smoothing theorem removes $\pi_2 = 0$ from the **compactification** conclusion. Bounded diameter, actual positive projection and simultaneous small Ricci/gauge defects yield a negative KE orbifold with a global orbifold complex structure. Smooth ALE blow-up limits are anti-self-dual.

To obtain an exact metric on the original smooth type, the note gives a separate theorem: a global Q-Gorenstein smoothing, together with matching of each whole geometric filling to its Milnor fiber **with its boundary marking**, gives an oriented diffeomorphism to a smooth fiber. Canonical ampleness and Aubin–Yau then provide the required exact metric. The Lee–Park logarithmic-tangent obstruction criterion supplies a checkable global smoothing condition; nodal hypersurfaces give an explicit classical example. Merely classifying a terminal ALE bubble does not supply the whole filling or this matching.

This is an alternative to excluding every bubble. The raw geometric data prove the orbifold conclusion; the additional smoothing and marked-topology conditions remain additional, not hidden consequences of saturation.

5. The next mathematical decisions are now distinct

Mechanism	What is now available	What is not supplied
Cover gauge supply	Exact geometry on the base, independent of cover degree	Arbitrary descent of an upstairs Einstein metric
Weighted propagation	Volume-independent correction of distributed residuals	Inversion in the presence of kernels or uncontrolled growth
Collapsed rewinding	Product geometry on the maximal-rank bulk	Global deck matching, bad-region topology and full inversion
Singular compactification	KE orbifold and a marked smoothing realization criterion	Automatic QG smoothing and whole-filling identification

These results suggest two focused continuations, not another general route list. One is to classify the global transition and low-mode data of the product-type collapsed regions, where the positive-coercivity repair theorem does not apply. The other is to determine when the geometric fillings produced by actual near-saturation must meet the relative Milnor criterion, or to construct a counterexample. Either can change exact existence; merely shrinking an average error or classifying one more terminal bubble cannot substitute for it.

The retained diameter-free integral obstruction theory remains independent and valid under its own actual-input assumptions. None of these additions proves the unrestricted obstruction/existence dichotomy or completes a Yau26 sign branch. All proofs, classical dependencies, exact model tests and local commits remain in this project; inherited sources and live systems are unchanged.